Exercise 2 : SQL Aggregate functions and SQL Operators.

Table: Students

student-id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

1. List all distinct departments in the student table.

```
SELECT DISTINCT department
FROM students;
```

OUTCOME

department
IT
HR
finance.

2. Get the average age of students per department.

Columns : department, avg-age

```
SELECT department
AVG (age) as avg-age
FROM students
GROUPED BY DEPARTMENT;
```

department	avg - age
IT	20.5
HR	22
finance	23

3. Show departments with more than 1 student.

Exp Colm : department, student - count

SELECT (student_id) AS student - count

FROM students

GROUPED BY department

HAVING student - count > 1;

department	student - count
IT	2
HR	2

4. Get all students whose age is between 21 and 23

exp colm : student-id, name, age, department.

SELECT student-id, name, age, department

FROM students

WHERE age BETWEEN 21 AND 23;

student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

5. List all students in the IT or HR department who are older than 21.

Exp colum: student-id, name, age, department.

SELECT student-id, name, age, department

FROM students

```
WHERE department IN ("IT', 'HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
4	Diana	23	finance
5	Eve	22	HR

Table : courses

course_id	course_name	department	credits
101	SQL. Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	finance	2
105	Statistics	HR	3

6. Show total credits per department, only for department with more than 5 total credits.

Exp columns: department, total credits.

```
SELECT department
sum (credits) AS total_credits
FROM courses
GROUPED BY department
HAVING sum (credits) > 5
```

department	total credits
IT	11

7. List all courses that do not have 4 credits.

Exp colm: course-id, course-name, department, credits.

SELECT course-id, course-name, department, credits

FROM courses

WHERE credits < > 4

course-id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Show the top 2 courses by credits in descending order.

Exp colm: course-id, course-name, credits.

SELECT course-id, course-name, credits

FROM courses

ORDER BY credits. DESC

LIMIT 3;

course-id	course-name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Table: enrollments

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

9. Get the maximum, minimum and avarage grade across all enrollments.

Exp colm: max-grage, min-grade, avg-grade

```
SELECT MIN (grade) AS min-grade
FROM enrollments;
```

min-grade
78

```
SELECT MAX (grade) AS max-grade
FROM enrollment;
```

max-grade
90

```
SELECT AVG (grade) AS avg-grade
FROM enrollments;
```

avg-grade
84.6

10. Count how many enrollments exist per course:

• course_id, enrollment_count.

```
SELECT COUNT (course_id) AS enrollment_count
FROM enrollments
GROUPED BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Table: salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

11. Find the total salary and total bonus per department

• department, total_salary, total_bonus

```
SELECT department,
    sum (salary) AS total_salary,
    sum (bonus) AS total bonus
    FROM salaries
  GROUBED BY department;
```

department	Total Salaries	Total bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. Show departments where avarage salary is above 55 000

- department, avg-salary.

SELECT department, AVG (salary) AS avg-salary
FROM salaries
Grouped by departments
HAVING avg-salary > 55 000

department	avg-salary
IT	61 000
Finance	70 000

13. List employees whose salary plus bonus is greater than 60 000.

- employee-id, name, salary, bonus, total compensation

SELECT employee-id, name, salary, bonus,
(salary + bonus) AS total-compensation
FROM salaries
WHERE total-compensation > 60 000

employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

Table : Projects

project - id	project - name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
5	HR Portal	HR	50 000

14. Show total and avg budget per department. Only include department with avarge budget above 70K.

• department, total-budget, avg-budget

```
SELECT department, sum (budget) AS total-budget,
    AVG (budget) AS avg-budget
    FROM projects
    Grouped by department
    HAVING AVG (budget) > 70 000;
```

department	total_budget	avg - budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budget between 50 000 and 120 000, excluding the marketing department.

• project - id, project - name, department, budget

```
SELECT project-id, project-name, department, budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department <> 'Marketing';
```

project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000